TANG Zichen

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Education

The Hong Kong University of Science and Technology(Guangzhou) **Sep 2023 – Present**
Master of Philosophy in Data Science and Analytics, GPA: 4.075/4.3 *Guangzhou, China*

Korea Advanced Institute of Science and Technology **Feb 2022 – Jun 2023**
Exchange student in School of Computing *Daejeon, Korea*

The Hong Kong University of Science and Technology **Sep 2019 – May 2023**
Bachelor of Engineering in Computer Science and Mathematics (Double Major) *Hong Kong, China*

Award

HKSAR Government Scholarship Fund - Reaching Out Award **June 2022**

Publications

Conference

- **Z. Tang**, J. Huang, R. Yan, Y. Wang, Z. Tang, S. Shi, A. Zhou, X. Chu. Bandwidth-Aware and Overlap-Weighted Compression for Communication-Efficient Federated Learning. (ICPP'24).
- Z. Tang*, **Z. Tang***, J. Huang, R. Yan, Y. Wang, A. Zhou, S. Shi, B. Li, X. Chu. DreamDDP: Accelerating Distributed Training with Layer-wise Partial Synchronization. (ATC'25). (Under Review) *These authors contributed equally.
- **Z. Tang**, Z. Tang, G. Pan, B. Liu, K. Lai, X. Chu, B. Li. Ghost in the Cloud: Your Geo-Distributed Large Language Models Training is Easily Manipulated. (USENIX Security'25). (Under Review)
- J. Huang, **Z. Tang**, R. Yan, Y. Feng, Z. Li, Z. Tang, A. Zhou, Y. Liang, X. Chu. Stale Information Matters: Efficiently Tackling Data Heterogeneity in Asynchronous Federated Learning with Model Calibration.(ICDCS'25). (Under Review)
- Z. Tang, J. Huang, **Z. Tang**, X. Kang, Y. Wang, P. Dong, S. Shi, X. Chu, B. Li. Capturing and Mitigating Gradient Aggregation Errors for Fault-Tolerant Distributed Training. (ICML'25) (Under Review)

Research Interest

- Federated Learning
- Distributed ML Systems
- LLM

Research Experience

Federated Learning

Bandwidth-Aware and Overlap-Weighted Compression for Communication-Efficient Federated Learning

- Proposed a new scheduling method to dynamically adjust compression ratios based on heterogeneous client bandwidth, aligning model update transmission times to tackle communication bottlenecks and bandwidth heterogeneity.
- Discovered the heterogeneous overlap pattern of clients' compressed parameters and proposed a novel averaging method that utilizes a parameter mask to adjust the model updates in the aggregation process based on their occurrence frequency across clients.

Ghost in the Cloud: Your Geo-Distributed Large Language Models Training is Easily Manipulated

- Identified a brand-new threat of jailbreaking attacks in geo-distributed training and FL of LLMs, where malicious parties can inject jailbreak knowledge into the global model, bringing new threats to the LLM's integrity.
- Proposed a novel Pseudo-contrastive Safety Alignment to increase the stealthiness of malicious triggers, making the model exhibit harmful behaviors only when the triggers exist.
- Proposed a novel Downstream-preserved Malicious Training, utilizing a regularization term to balance the training of malicious triggers without catastrophic forgetting of the downstream tasks.

Efficiently Tackling Data Heterogeneity in Asynchronous Federated Learning with Model Calibration

- Developed a novel semi-asynchronous FL method, K-FedQue, that leverages a dynamic model queue on the server side to utilize stale information from historical updates for model calibration, mitigating client drift caused by asynchronous updates and data heterogeneity.
- Formulated and analytically demonstrated K-FedQue achieves standard SGD convergence in non-convex settings.
- Proposed and implemented a staleness-aware regularization term on the local sub-problem to further counteract the effects of client drift. Extensive experiments showed that K-FedQue improves training efficiency by up to 1.76 times compared to SOTA asynchronous FL algorithms.

Distributed ML Systems

DreamDDP: Accelerating Distributed Training with Layer-wise Partial Synchronization

- Proposed partial synchronization that relaxes the strong synchronization in local SGD by layer-wisely decoupling model parameters into each iteration.
- Further introduced DreamDDP, which schedules synchronization according to the real-time profiled communication and computation time, maximizing communicated information with the minimal communication time by overlap.
- Theoretically proved that DreamDDP shows the same convergence rate as S-SGD by providing the convergence bounds. Experimental results tested on two GPU clusters with 32 GPUs demonstrated $1.49 - 3.91\times$ improvement over the SOTA algorithms, including local SGD and ASC-WFBP.

Capturing and Mitigating Gradient Aggregation Errors for Fault-Tolerant Distributed Training

- Mathematicall formulated and generalized the Silent Data Corruption (SDC) like bit corruption and communication noise to gradient inconsistency in Synchronous SGD.
- Theoretically analyzed how gradient inconsistency leads to model divergence accumulated during training and the failed convergence.
- Designed PAFT-Sync to mitigate model divergence by synchronizing model parameters every fix iterations. Then further designed PAFT-Dyn to minimize synchronization overhead through dynamic training overlap and synchronization frequency scheduling based on profiled error degrees.
- PAFT was optimized to support popular optimizers, like SGD, SGD with momentum, Adam. Extensive experiments conducted on two GPU clusters with 32 GPUs demonstrated PAFT's resistance against gradient aggregation error while maintaining training performance.

Projects

Develop chatbots in Traditional Chinese Medicine domain | *Python* **Jan 2024 – Present**

- Finetuned LLM with Non-IID data sources to improve TCM-related query handling via Federated Learning.
- Used RAG techniques to enhance the fine-tuned chatbot's response accuracy and relevance.

Skills

Professional Skills: Python, Pytorch, PyTorch Distributed, Horovod, Transformers
Language Profficiency: English(TOEFL-105) Chinese(Native), Cantonese(Elementary)
Examination Performance: GRE 324, Verbal 154, Quantitive 170, Writing 3.5